MORGAN R. PFAFF

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EDUCATION

University of California (UC) Berkeley and UC San Francisco **Berkeley, CA**
Doctor of Philosophy, UC Berkeley–UCSF Joint Graduate Program in Bioengineering October 2026

- Dissertation: "Semi-synthetic Hyaluronic Acid Hydrogels to Direct Fibro-Adipogenic Progenitor Fate in Injured Rotator Cuff Muscle"; Advisor: Dr. Kevin Healy
- Major: Biomaterials Engineering & Tissue Regeneration; Minor: Cell, Tissue, & Developmental Biology
- GPA 4.0/4.0; National Science Foundation Graduate Research Fellow

Northeastern University **Boston, MA**
Bachelor of Science in Biochemistry May 2021

- GPA 4.0/4.0; Honors Program; Honors in the Discipline
- Engineering & entrepreneurship study abroad program, Oxford, United Kingdom (Summer 2019)

RESEARCH EXPERIENCE

Kevin Healy Lab, UC Berkeley **Berkeley, CA**
PhD Candidate January 2022 – October 2026

- Engineered viscoelastic hyaluronic acid (HyA) hydrogels for rotator cuff muscle regeneration; characterized rheology, injectability, and chemical composition of different formulations; and developed high-throughput PDMS culture arrays that enabled small-scale 3D testing of hydrogels with precious patient-derived cells
- Analyzed single-cell transcriptomes of patient-derived fibro-adipogenic progenitors (FAPs) across four induction states to identify adhesion receptors and matrix-remodeling enzymes they use to sense and reshape their surroundings, and designed formulations that tuned ligand density and degradability independently of stiffness
- Demonstrated that the hydrogel shifts FAPs toward a pro-regenerative beige phenotype and studied its effects on muscle stem cell fusion in 3D cultures and cocultures
- Revived and updated historical lab methods, improved hydrogel fabrication and characterization SOPs, and helped coordinate reagent synthesis, inventory, and quality control
- Prepared hydrogels for collaborators' *in vivo* studies at the University of Virginia, UCSF, and partner labs, and helped plan experiments and interpret results to guide next-generation designs, including a mouse study in which the hydrogel alone reduced fibrosis and fat infiltration
- Published 2 peer-reviewed papers (1 first-author) and 3 in preparation, gave 5 presentations at 4 national and international conferences, and contributed data and writing to ~6 grant proposals, including an NIH R01

Randall Erb Lab, Northeastern University **Boston, MA**
Undergraduate Researcher September 2016 – July 2021

- Developed dual-network hydrogels reinforced with magnetically responsive hydroxyapatite fibers for bone grafts, characterizing their mechanics and microstructure (dynamic mechanical analysis, SEM) and modeling fiber rotation as a potential toughening mechanism with COMSOL finite element simulations
- Fabricated freeze-cast collagen conduits with aligned, double-layered porosity for peripheral nerve repair in collaboration with Dartmouth College and Harvard Medical School

Marc Hansen Lab, University of Connecticut Health Center **Farmington, CT**
Summer Research Fellow June – August 2017; June – August 2018

- Piloted synthetic-lethal inhibitor and siRNA combinations in osteosarcoma and hFOB cells to find druggable vulnerabilities in growth factor receptor signaling, providing preliminary data for NIH grant applications

- Used laser capture microdissection on histological sections to isolate osteoblast and osteoclast populations and investigate cell behavior and genetics in Paget's disease of bone

TRANSLATIONAL, REGULATORY, & INDUSTRY EXPERIENCE

MuscleMatrix **Berkeley, CA**
Scientific Consultant August 2026 – Present

- Consult for an early-stage startup, spun out in part from doctoral research, that is developing an injectable, cell-free hyaluronic acid scaffold for muscle regeneration after rotator cuff injury; named co-inventor of its technology
- Shape FDA regulatory strategy for a medical device or combination product, and write grant applications
- Build company operations from the ground up, including regulatory and environmental health and safety (EHS) compliance, lab buildout, equipment procurement, manufacturer and supplier sourcing, and staff training
- Conduct formulation development, characterizing how drug loading affects hydrogel injectability and mechanics to inform a planned drug-eluting version

Vertex Pharmaceuticals **Boston, MA**
Exploratory Research Co-op January 2020 – June 2020

- Developed and optimized a reporter assay to measure transcription factor activity in potentially druggable sickle cell disease (SCD) pathways, giving the team a faster, more accurate readout of compound effectiveness, and built the stable cell lines it required by molecular cloning
- Independently planned, executed, and analyzed experiments, and presented results and journal club articles to the SCD exploratory research team

Editas Medicine **Cambridge, MA**
Screening Group Co-op January 2018 – June 2018

- Screened sgRNAs to identify the most effective candidates for the company's pipeline diseases, introducing CRISPR-mediated edits via nucleofection, transfection, and viral transduction
- Programmed and optimized high-throughput screening assays on automated liquid handlers, enabling more complex, higher-throughput sgRNA screening; analyzed data in Prism, Tableau, and FlowJo and presented findings to the screening group and other departments

TEACHING & MENTORING

University of California, Berkeley **Berkeley, CA**
BIOE/MSE C118/C208: Biological Performance of Materials August 2022 – Present

- Graduate Student Instructor (Fall 2022; 45 students), Reader (Fall 2026; 30 students), and Guest Lecturer (2023–2026) for Dr. Kevin Healy's course
- Delivered 4 full-length lectures (2 on hydrogels, 1 on polymer viscoelasticity, and 1 introduction to polymers)
- Led weekly discussion sections and office hours, and evaluated ~10 team design projects, including 15-page proposals for biomaterial solutions to unmet clinical needs and pitch, 10-, and 20-minute presentations
- Form student design teams and grade homework, exams, written reports, and presentations

BIOE 252: Clinical Need-Based Therapy Solutions August 2026 – Present

- Reader for Dr. Syed Hossainy's course (~100 students): guide teams developing therapy concepts for unmet clinical needs, grade coursework, and coordinate guest lecturers from around the country

Research Mentor, Kevin Healy Lab January 2022 – October 2026

- Undergraduate researcher (2023–2024): integrated hydrogels into a microphysiological system to study FAP migration in a physiologically relevant model; now a PhD student at Johns Hopkins University
- Master of Translational Medicine capstone team (4 students, 2024–2025): characterized hydrogel rheology to assess injectability and conducted a market analysis for rotator cuff injury indications
- Visiting PhD student from Spain (2024): fabricated carbon nanotube–HyA hydrogel composites, tested their conductivity, and assessed the beating behavior of iPSC-derived cardiomyocytes encapsulated in them

- 3 first-year PhD rotation students (2022–2026): developed a degradation-based workflow to isolate cells from HyA hydrogels, optimized rheology workflows to prevent sample evaporation during long-duration tests, and built machine learning–based screening workflows for earlier identification of FAP fate
- Trained dozens of additional students and lab members on equipment and methods

PUBLICATIONS & PRESENTATIONS

Publications

1. Wang, V., Zhu, Y., Spaulding, K., **Pfaff, M.**, Neiman, G., Takaba, K., Chen, N., Parekh, K., Hasaballa, A., Haraldsson, H., Baker, A., Saloner, D., Wallace, A., Ziats, N., Healy, K., Ratcliffe, M. NIPAAm-co-PEG4000 thermoresponsive hydrogel with ROS scavenging properties improves in vivo border zone contractility and reduces myocardial remodeling in sheep after MI. *Acta Biomaterialia*, 2026, 211, 48–60. doi.org/10.1016/j.actbio.2025.06.046
2. **Pfaff, M.**, Wague, A., Davies, M., Killaars, A., Ning, D., Garcia, S., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B., Healy, K. Viscoelastic HyA hydrogel promotes recovery of muscle quality and vascularization in a murine model of delayed rotator cuff repair. *Advanced Healthcare Materials*, 2025, 14, 2403962. doi.org/10.1002/adhm.202403962

Manuscripts in Preparation

1. **Pfaff, M.**, Chen, N., Jilani, H., Garcia, S., Lau, J., Wague, A., Hohman, O., Liu, X., Feeley, B., Healy, K. Transcriptome-guided design of hyaluronic acid hydrogels to direct human fibro-adipogenic progenitor fate. In preparation.
2. **Pfaff, M.**, Sagar, R., Andrade, B., Hines Ramirez, D., Pal, S., Goswami, I., Messersmith, P., Healy, K. Bioinspired adhesive for reversibly bonded microphysiological systems. In preparation for *Lab on a Chip*.
3. Wague, A., **Pfaff, M.**, Fusagawa, H., Lau, J., Sharma, S., Youn, A., Healy, K., Liu, X., Feeley, B. Hydrogel-delivered beige adipose precursors enhance muscle quality and functional recovery in a delayed rotator cuff repair mouse model. In preparation for *Cell Biomaterials*.

Oral Presentations

1. **Pfaff, M.**, Wague, A., Davies, M., Killaars, A., Ning, D., Garcia, S., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B., Healy, K. Viscoelastic HyA hydrogel promotes recovery of muscle quality and vascularization in a murine model of delayed rotator cuff repair. Presented at: Biological Scaffolds for Regenerative Medicine (BSRM); May 2025; Napa, CA.
2. **Pfaff, M.**, Killaars, A., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B., Healy, K. Tunable semi-synthetic hyaluronic acid hydrogels promote muscle regeneration after rotator cuff injury. Presented at: Biomedical Engineering Society (BMES) Annual Meeting; October 2023; Seattle, WA.
3. **Pfaff, M.**, Killaars, A., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B., Healy, K. Semi-synthetic hyaluronic acid hydrogels promote muscle regeneration by inducing BAT differentiation of FAPs. Presented at: Society for Biomaterials (SFB) Annual Meeting & Exposition, "Biomaterials for Regenerative Engineering"; April 2023; San Diego, CA.
4. **Pfaff, M.**, Faust, J., Erb, R. Dual-network hydrogels reinforced with field-responsive hydroxyapatite exhibiting high strength and stiffness while fully hydrated. Presented at: Materials Research Society 2020 Spring/Fall Meeting and Exhibit; November 2020; Virtual.
5. **Pfaff, M.**, Faust, J., Erb, R. Dual-network hydrogels reinforced with field-responsive hydroxyapatite exhibiting high strength and stiffness while fully hydrated. Presented at: Northeastern Regional Student Conference of the Society for Experimental Mechanics; June 2020; Virtual.

Poster Presentations

1. **Pfaff, M.**, Killaars, A., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B., Healy, K. Tunable semi-synthetic hyaluronic acid hydrogels promote muscle regeneration after rotator cuff injury. Presented at: Orthopaedic Research Society (ORS) 50th International Musculoskeletal Biology Workshop; July 2023; Midway, UT.
2. **Pfaff, M.**, Killaars, A., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B., Healy, K. Semi-synthetic hyaluronic acid hydrogels promote muscle regeneration by inducing BAT differentiation of FAPs. Presented at: Society for Biomaterials (SFB) Annual Meeting & Exposition; April 2023; San Diego, CA.

3. **Pfaff, M.**, Erb, R. Understanding rod rotation in fiber reinforced hydrogels: a potential toughening mechanism. Presented at: Research, Innovation, and Scholarship Expo; April 2021; Virtual.
4. **Pfaff, M.**, Alander, C., Hansen, M. Synthetic lethal approaches to treating osteosarcoma. Presented at: University of Connecticut Health Center Summer Research Fellowship Program; July 2017; Farmington, CT.
5. **Pfaff, M.**, Faust, J., Erb, R. Dual-network hydrogels with functionalized CaP rods for bone graft applications. Presented at: Research, Innovation, and Scholarship Expo; April 2017; Boston, MA.
6. **Pfaff, M.**, Faust, J., Erb, R. Dual-network hydrogels with functionalized CaP rods for bone graft applications. Presented at: National Conference on Undergraduate Research; April 2017; Memphis, TN.

FELLOWSHIPS & AWARDS

National Science Foundation Graduate Research Fellowship ($138,000)	2021
Research, Innovation, and Scholarship Expo (RISE) Excellence in Research Award ($2,000)	2021
PEAK Summit Award, undergraduate research grant ($3,000)	2020
PEAK "Shout It Out!" Award, conference travel grant	2020
RISE Innovation Alley Selection (poster selected for highest impact and commercial potential)	2017

LEADERSHIP & SERVICE

Food Not Bombs East Bay **September 2024 – Present**

- Lead a weekly shift, picking up meals prepared by volunteer cooks and serving 25–40 free vegetarian meals to unhoused and low-income community members
- Transformed an unstaffed curbside drop-off into an in-person meal service, creating a more personal experience and building ongoing relationships with community members

Sigma Xi, The Scientific Research Honor Society, UC Berkeley Chapter **2021 – Present**

Vice President 2023 – Present

- Founded (2021) and chair the chapter's Committee for Student Advancement, growing student involvement from 2 to ~35 undergraduate and graduate students; lead a 3–6-person board that meets weekly to biweekly
- Mentored 17 students through Sigma Xi membership nominations and oversaw a board-run voluntary mentorship program pairing undergraduate pre-members with associate members
- Organized 20+ chapter events, including a 5-part student and postdoc seminar series, 11 community-building socials, 4 new-member info sessions, and an induction ceremony; hosted the Editor-in-Chief of *American Scientist* for a Faculty Club lecture and dinner
- Independently organized the Sigma Xi–ATP Bio Research Symposium & Poster Session, from securing a funding partner to day-of logistics, drawing ~25 presenters (including students from nearby universities and community colleges), a keynote speaker, and ~150 attendees
- Co-organize the semi-annual Grants in Aid of Research program, awarding up to $750 to undergraduate, master's, and PhD students across STEM, and lead chapter recruitment through campus groups and departments

Sigma Xi, Northeastern University Chapter **2020 – 2021**

Diversity & Inclusion Chair

- Expanded outreach to first-generation, culture-specific, and other underrepresented student organizations, and hosted a campus-wide diversity, equity, and inclusion panel

BioEngineering High School Competition (BioEHSC) **April 2022 – April 2026**

- Academic Judge: evaluated 10–15 high school team pitches each spring in a full-day bioscience pitch competition, assessing depth of knowledge through Q&A and providing feedback

Northeastern University Service & Leadership **September 2016 – June 2021**

- Triathlon Team President/Captain (2019–2020), Chief Financial Officer (2018–2019), and Chief Procurement Officer (2017–2018); Club Sports Council liaison between teams and administration (2019–2021)

- Resident Assistant (2017–2019): live-in mentor for ~45 first-year women in STEM each year, designing programs such as Q&A panels with graduate students and professors and faculty mentorship events
- Volunteer swim mentor for a Boston Children's Hospital patient with a disability through Adventure Challenge (2018–2019), and AVID college-readiness tutor at Fenway High School (2018)

CLINICAL EXPERIENCE

Tufts Medical Center, Oncology and Hematology **Boston, MA**
Clinical Care Technician January 2019 – December 2019
- Provided direct care for 6–10 oncology and hematology patients per shift, monitoring vital signs, escalating abnormal findings to nursing staff, and supporting clinical teams during procedures and emergencies

Fallon Ambulance Service **Quincy, MA**
Emergency Medical Technician (EMT-Basic) September 2018 – April 2019
- Assessed, treated, and transported ill and injured patients, providing immediate medical and psychological care en route to hospitals and specialty facilities

Dental Office of Dr. Andrew J. Ponichtera, DMD, MS **Simsbury, CT**
Dental Assistant June 2016 – August 2016
- Supported chairside procedures, including fillings, root canals, crowns, implants, bridges, and dentures

SKILLS

Biomaterials Synthesis & Mechanical Testing
- Hydrogel synthesis and chemical modification, drug loading and release, rheology, injectability testing, mechanical testing (dynamic mechanical analysis, tensile, burst, and lap shear), freeze casting, lyophilization

Materials Characterization & Microfabrication
- NMR spectroscopy, X-ray photoelectron spectroscopy (XPS), scanning electron microscopy (SEM), microfabrication and photolithography

Cell & Molecular Biology
- Primary and 3D cell culture, immunocytochemistry, confocal and fluorescence microscopy, histology, flow cytometry, qPCR/ddPCR, Western blot, molecular cloning, CRISPR-Cas9 gene editing (transfection, nucleofection, viral transduction), automated liquid handling

Data Analysis & Computation
- R and Python (incl. single-cell RNA-seq), GraphPad Prism, MATLAB, COMSOL (finite element analysis)